Unit III

Introduction to data Analysis and Algorithms

Data Analysis:- it the process of inspecting, cleaning, transforming and modeling data to extract useful information, for decision making.

Algorithm: are steps by step procedure used for calculations, data processing and automated reasoning.

Role of statistics in Data analysis:-

(1) Summarizing data (descriptive statistics)

(2) making inference from data (Inferential statistics)

(3) Identifying patterns and trends.

(4) Testing hypotheses.

Types of statistics

(1) Descriptive statistics! -

understanding, organizing, summarizing and storing

the data

It Helps us describe the dataset using:

- Central Tendency: mean, median, mode
- Dispersion/variation → range, variance, standard deviation, IQR, histogram, boxplot.

Exp. Test scores of 5 students = {50, 60, 70, 80, 90}

- mean = 70
- Standard Deviation = 14.14 (shows variation from mean)

(2) Inferential statistics (making predictions/ Generalizations)

- used to draw conclusion from a sample and apply them to the whole population.

Exp. survey: 200 voters → 120 support candidate A

support % = 60%.

∴ conclusion: candidate A may win with ~60%.

sport in the entire voting population.

Exp. one tiny dataset

$$D = [2, 4, 4, 4, 5, 5, 7, 9] \quad n = 8$$

Central Tendency and its measure (wrt to D)

(a) mean (average)

$$\bar{x} = \frac{2+4+4+4+5+5+7+9}{8} = \frac{40}{8} = 5.0$$

(b) median (middle)

Even n: average of 4^{th} & 5^{th} values

$$\frac{4+5}{2} = 4.5$$

useful :-

(c) Mode (most frequent)

Mode = 4 (appeals 3 times)

Useful : the most common value (great for categorical data).

(2) Variations (measure of spread) :-

(a) Range = max - min = 9-2 = 7

(b) Variance (average squared deviation from mean):

$$X = [2,4,4,4,5,5,7,9]$$

Mean (μ or $\bar{x}$) :-

$$\mu = \frac{2+4+4+4+5+5+7+9}{8}$$

$$= \frac{40}{8} = 5$$

(2) So, mean = 5.

-Us step (2) Deviations from the mean

ap subtract mean (5) from each data

Ex point :-

$$[2-5, 4-5, 4-5, 4-5, 5-5, 5-5, 7-5, 9-5]$$

Co $= [-3, -1, -1, -1, 0, 0, 2, 4]$

So deviations = $[-3, -1, -1, -1, 0, 0, 2, 4]$

(3)

Squared Deviations

$[-3^2, (-1)^2, (-1)^2, (-1)^2, (0)^2, (0)^2, (2)^2, (4)^2]$

$= [9, 1, 1, 1, 0, 0, 4, 16]$

Sum = 32

variance :- It is the average squared deviation from the mean.

There are two types:

① Population variance (σ^2)

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

Here, N = 8 (all value considered)

$$\sigma^2 = \frac{32}{8} = 4.0$$

Population variance = 4.0

Exp. T-test, Z-test, confidence Interval, hypothesis test,

(2) Sample Varience (S^2)

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Here, $n = 8$ (Sample size)

$$S^2 = \frac{32}{7} = 4.5714$$

Sample varience = 4.57

Standard deviation (SD) :-

SD is simply the square root of varience.

1. population standard deviation (σ)

$$\sigma = \sqrt{\sigma^2} = \sqrt{4.0} = 2$$

Sample standard deviation (s)

$$s = \sqrt{s^2} = \sqrt{4.5714} = 2.14$$

Sample SD = 2.14

Interpretation :-

- Varience tells us how spread out the data is (in squared unit)
- Standard deviation tells us the spread in the same unit as the data.

in our dataset :-

mean = 5

- Population SD = 2 → most data lies with in 2 unit of the mean.
- this mean value are generaly b/w 3 and 7.

What are Quartiles and IQR:-

Quartiles split ordered data into four equal parts

Q1 = 1st quartile (25th percentile)
- middle of the lower half.

Q2 = median (50th percentile)

Q3 = 3rd. quartile (75th percentile)
middle of the upper half.

- IQR (interQuartile Range) = $Q_3 - Q_1$

it measures the spread of the middle.
50% Robust to outliers

Method A

Steps (even n)

1. Find the median (Q2). For even n, median = average of the two middle values.
2. Split data into lower half (values before median) and upper half (values after median)
3. Q1 = median of the lower half,

   Q_3 = median of the upper half.

Let's see exp.

D = [2, 4, 4, 4, 5, 5, 7, 9] n=8 sorted

Apply to D: [2, 4, 4, 4, 5, 5, 7, 9]
1 2 3 4 5 6 7 8 → index

$n = 8 \rightarrow$ middle positions are 4 and 5.

$$Q_2 = \frac{4+5}{2} = 4.5$$

2. Lower half = first 4 values

: [2, 4, 4, 4]

upper half = last 4 values

[5, 5, 7, 9]

lower-half median (Q1): middle of [2, 4, 4, 4]

positions 2 & 3 $\rightarrow (4+4)/2 = 4$

upper-half median (Q3): middle of [5, 5, 7, 9]

$\rightarrow (5+7)/2 = 6$

Ex:(2) if n is odd (n=9)

$$X = \{3, 5, 7, 8, 10, 12, 13, 15, 18\}$$

Step (2) Find the median (Q_2)

median is the middle value.

since $n = 9$, (odd), the median is at position $\frac{(n+1)}{2} = \frac{(9+1)}{2} = 5$.

So, $Q_2 = 10$ (5th value)

step (3) split data into lower half and upper half

— important rule for odd n:

Do not include the median in either half

lower half (first 4 values): $\{3, 5, 7, 8\}$

upper half (last 4 values): {12, 13, 15, 18}

step (4)

Find Q_1 (lower Quartile)

Q_1 is the median of the lower half.

lower half = {3, 5, 7, 8}, size = 4 (even)

median of 4 numbers = average of 2 middle values

$$Q_1 = \frac{5+7}{2} = 6$$

step (5) Find Q_3 (upper Quartile)

Q_3 is the median of the upper half

upper half = {12, 13, 15, 18}

median = average of 2 middle values

$$Q_3 = \frac{13+15}{2} = 14.$$

Step (6) Calculate interquartile range (IQR)

$= Q_3 - Q_1 = 14 - 6 = 8$

Final Result (n = 9)

$Q_1 = 6$

Q_2 (median) $= 10$

$Q_3 = 14$

IQR = 8.

Result (method A):-

$Q1 = 4,\ Q2 = 4.5,\ Q3 = 6$

$IQR = 6 - 4 = 2$

Outlier detection using IQR :-

Lower fence $= Q1 - 1.5 * IQR$

upper fence $= Q_3 + 1.5 * IQR$

Points outside these fences are commonly flagged as outliers.

Method A $IQR = 2$

Lower $= 4 - 1.5 * 2 = 4 - 3 = 1$

Upper $= 6 + 3 = 9.$

Data points outside (<1 and >9) $\rightarrow 9$

worked. example for an odd-size list

$Y = [1,2,3,4,5]$ $(n = 5)$

median $Q_2 = 3$.

median of halves (exclude median):

lower half $= [1,2] \rightarrow Q_1 = (1+2)/2 =$

1.5

upper half $= [4,5] \rightarrow Q_3 = 4+5/2 = 4.5$

finally Repeat the steps.